

Nested-quotient temporal histories and homological flat bands in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Final edition, with machine-checked provenance

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. After selecting the complete gauge-invariant charge-odd one-plaquette eigenspace, fixed-order contributions are organised as ordered magnetic histories on the Haar–Gram quotient of reachable Wilson networks. What is exact and all-rank is one matrix element: the connected second-order coupling between two plaquettes sharing an oriented link is $t_N u^2$ times their incidence overlap, with

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

Under two stated hypotheses — that the retained shell is exactly the charge-odd one-plaquette span, and that no connected non-adjacent history carries momentum dependence — those elements assemble into the oriented edge–face Gram operator $t_N u^2 B(k)B(k)^\dagger$ with $B = \partial_2^\dagger$, whose projected spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, $q = 4 \sum_j \sin^2(k_j/2)$. Neither hypothesis is checked here and both are named where they are used. The flat carrier is $\ker \partial_2$, of exact dimension $L^3 + 2$, and that is unconditional.

Relative to the 28 August edition this one adds five things, each carrying its own machine check. The channel decomposition of t_N , previously a representation-theoretic ledger derived on a single plaquette pair, is recovered on a genuine face-sharing two-cube $SU(3)$ cluster — an exact-Casimir $B = 6$ basis of 1,590,462 states, which the delivery reaches by path enumeration without ever assembling the magnetic matrix — and the two decompositions agree *channel by channel* rather than only in the sum (Reported result 14); on that space each of the six channels is separately proportional to the same incidence Gram, on all 56 adjacent pairs, so the separation of colour from geometry is measured and not assumed, and the connected diagonal — eleven numbers in

both deliveries — is constant on the prism’s three symmetry orbits, with the whole $B = 4 \rightarrow B = 6$ difference confined to the eight faces that carry connected transport (Proposition 18). Separately, and by arithmetic on the weight ledger alone, a cutoff keeping only the singlet and $\Lambda^2 F$ routes projects the hopping to $-1/12$ — opposite in sign to t_3 — and omits exactly 14/153, the two summing to t_3 (Corollary 16); the $B = 4$ truncation of the two-cube space delivers the first of those two numbers outright, on the same geometry. The charge-even sector acquires a closed form: with $\hat{a}_m = 2 + 2 \cos k_m$ and $p = \sum_m \hat{a}_m = 12 - q$, its Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$, so one zone function runs both sectors (Theorem 21), and its Γ -point splitting pins the charge-even hopping $t_{r,+}$ — a separation the charge-odd rest frame has no analogue for, since its Γ spectrum has only one level (Proposition 24). The homological reading of the flat direction is shown to be the same one on a second complex: the six faces of a cube are a closed surface, its $\ker \partial_2$ is the fundamental class, and the one-cube charge-odd shell splits as $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$ with the flat level sitting at the scalar for *every* value of the hopping (Proposition 19). Finally the open fourth-order off-axis coefficient is given a geography rather than a verdict: an exact obstruction certificate, an explicit second witness, and the measurement that no Γ -anchoring error can move it at all (Section 9).

The two-cube channel census, and the $SU(3)$ third- and fourth-order statements, remain conditional on supplied ledgers — the first on six rational reconstructions of finite-precision contractions, the others on exact-arithmetic ledgers — whose raw payloads are not reproduced here, and all are labelled as such throughout. Every displayed result carries the name of the machine check that establishes it. All statements concern a formal finite-order projected sector: this is not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [1, 2]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology, line-graph constructions, compact localised states, and the need for noncontractible states on periodic systems are all established, and closely related closed-surface structures occur in two-form spin liquids. A recent classification of exactly flat bands protected by local graph topology [21] reaches extensive face-graph degeneracy from an *unoriented* face-vertex incidence, where rank-nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face-edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal $L^3 + 2$ — homology, not counting. It is recorded as a distinction, not an identification. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

check: the discrete index theorem gives 0 on the signed face-edge operator, bounding nothing

The content is dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an oriented edge-face Gram operator in the charge-odd one-plaquette sector. Representation theory fixes its coefficient and the cubical chain complex fixes its kernel and spectrum. The separation is exact,

$$\begin{aligned} \text{colour dynamics} &\longrightarrow t_N, \\ \text{cellular geometry} &\longrightarrow BB^\dagger. \end{aligned}$$

What is proved, and what is conditional

Nine statements are self-contained here.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2 , derived from the order-two Weingarten values (Theorem 7). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.
2. For every $N \geq 3$ the connected second-order coupling between two plaquettes sharing one link is

$t_N u^2$ times their incidence overlap (Theorem 10), and the channel list it runs through is complete (Lemma 6). The assembly of those elements into the torus operator is *not* here; it is item 10 below.

3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 1), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 2). The two are the same statement, by a sum rule over the momentum grid (Theorem 4) — and not merely in agreement: the Γ block is the harmonic subspace, so its 3 is $b_2(T^3)$, and the wrapping sheets averaged over their translates (8) are harmonic representatives of the three classes.
4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_1 = -1/12$, and what it omits is $w_{\text{Sym}^2 F} - w_{\text{Adj}} = 14/153$; the two sum to t_3 (Corollary 16). This is arithmetic on (12) and needs no input beyond Theorem 7.
5. The charge-even Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 21); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 22); and its Γ -point splitting is $12t_{r,+}$ exactly (Proposition 24). The identity is self-contained; the two numbers it is evaluated at inherit the status of $t_{r,+}$, and the $r = 3$ one is conditional — see item 12. That the charge-odd rest frame is blind to the hopping and the charge-even one is not follows from Schur’s lemma alone: the charge-odd triple carries the irreducible T_1 and the charge-even one $A_1 \oplus E$, so the two Γ blocks have one level and two, at every order and independently of the incidence ansatz.
6. The vacuum-mediated route $p \rightarrow |0\rangle \rightarrow p'$ connects any pair of plaquettes and is worth $1/C_F$; it vanishes in the charge-odd sector because V is charge-even, which is what makes Theorem 10’s shared-link restriction true here and false in general, and it gives the all-rank charge-even hopping ℓ_N in closed form (Proposition 12).
7. A correction of the boundary-factorised form $a_r I + b_r S + B M_r B^\dagger$ shifts the carrier without dispersing it, for *every* M_r (Proposition 28). The protection is a property of the boundary operator, not of the dynamics — and the proposition says nothing about a correction that does not factor through links, which is why the fourth order has to be argued.
8. On the closed cube surface the charge-odd shell is $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$, with G -eigenvalues $0, 4, 6$; the labels follow from the rotation action rather than being assigned, the A_1^{--} level is the fundamental class of S^2 , and it sits at the scalar for every hopping coefficient (Proposition 19). This is integer

linear algebra and takes no input from any delivery.

9. The retained Γ /axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 29), and an explicit second witness exhibits the non-identifiability by construction (Proposition 31).

Two families of statements are *conditional* on supplied exact-arithmetic derivation ledgers. Their raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not included here, and repeated assertions in synthesis documents are not treated as independent reproductions.

10. On two hypotheses that no check covers — that the range of P_- is exactly the charge-odd one-plaquette span, and that no connected non-adjacent second-order history carries momentum dependence — the adjacent elements of item 2 assemble into $t_N u^2 B B^\dagger$, with the band structure that follows from it (Conditional corollary 11). Everything downstream that quotes a bandwidth or a band edge at order u^2 inherits those two hypotheses.
11. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 14) — and the $B = 4$ truncation of that space retains exactly the adjacent shared-link channels the published link cutoff of item 4 retains, so the two land on the same projected hopping — the same channels, not the same scheme. Conditional on the delivered diagonals, the symmetry analysis of Proposition 18 applies to them: the orbit structure itself is exact and takes no delivered input. The $B = 6$ two-cube build was not reproduced here.
12. For $SU(3)$, supplied ledgers report that all candidate third-order lifter classes vanish and report the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, and nothing here checks any of them. If it is exhaustive and the linked ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3 .
13. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \dots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4 . The

complete off-axis operator remains open — and by Theorem 29 it cannot be closed by any Γ or axial datum.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2} . Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_\ell C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \text{Re Tr } U_p\right). \quad (1)$$

Dropping the additive constant,

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

with $\chi_p = \text{Tr } U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$ -specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u . Reading it as a coordinate would demand dividing the order- r coefficient by 4^r , which misses the printed tables by exactly 16 at u^2 and 64 at u^3 .

check: the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim

check: the 4**r rescaling breaks the bridge: order 2 off by 16, order 3 by 64

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (3)$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\left(\frac{1}{2}C_F\right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

This external energy is the input to the resolvent denominators of Section 4, where it is checked together with them. For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The charge-even hopping does *not* vanish there, which locates the $N^2 - 4$ zero in the channel difference rather than in the incidence.

check: at $N = 2$ the C-odd hopping vanishes and the C-even one does not

We work inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric centre flux around each torus cycle. Let P_- be the full $E_{F,N}$ spectral projector there. For $L \geq 3$ its range is the charge-odd one-plaquette span, since a reduced contractible fundamental loop at that energy has four distinct links and is an elementary plaquette. This sector qualification also excludes the accidentally degenerate straight wrapping loop at $L = 4$. With $Q = 1_- - P_-$ the reduced resolvent crosses no omitted $E_{F,N}$ eigendirection, and degenerate perturbation theory gives

$$H_{\text{eff},-} = P_- H_E P_- + u H_1 + u^2 H_2 + u^3 H_3 + \cdots \quad (5)$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q -space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2 \partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1 \partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4 \sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (6)$$

Theorem 1 (Incidence factorisation). *With $w(k) = (\bar{d}_3, -\bar{d}_2, \bar{d}_1)^\top$,*

$$BB^\dagger = qI - ww^\dagger, \quad B^\dagger w = 0, \quad \|w\|^2 = q,$$

and therefore $\text{spec}(BB^\dagger) = \{0, q, q\}$. With $S = BB^\dagger - 4I$ the signed shared-link adjacency, $\text{spec } S(k) = \{-4, q-4, q-4\}$.

Proof. Multiplying (6) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its

orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. $\square$

check: $B B^\dagger = q I - d \text{ conj}(d)^\top$ for the curl incidence

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ , $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis [20].

Theorem 2 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 , $\ker \partial_2 = \text{im } \partial_3 \oplus H_2$ with $\dim \text{im } \partial_3 = L^3 - 1$ and $\dim H_2 = 3$, so $\dim \ker \partial_2 = L^3 + 2$.*

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\text{rank } \partial_3 = L^3 - 1$ by rank-nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. $\square$

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \dots, 5$.

check: $\dim \mathbb{Z}_2 = L^3 + 2$ by rank, not by re-arranging the formula

check: cube boundaries and three wrapping sheets SPAN $\mathbb{Z}_2$

Remark 3. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x+e$ and $x-e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

check: the L^3+2 count is chain-level, not the Bloch convention

The following sum rule joins Theorems 1 and 2, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$ -dimensional space.

Theorem 4 (Bloch-chain bridge). *On the periodic L^3 complex,*

$$\sum_k (3 - \text{rank } B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other $L^3 - 1$ momenta.

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 1 gives $\text{rank } B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. $\square$

check: the Bloch and chain routes to the carrier agree

Remark 5. Earlier editions of this paper read the agreement as a coincidence worth checking — “the 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$ ”. That was wrong, and the correction is the better statement. Under the discrete Fourier transform ∂_2 block-diagonalises and $\ker \partial_2 = \bigoplus_k \ker \partial_2(k)$; at $k \neq 0$ the kernel block is one-dimensional and is exactly $\text{im } \partial_3(k)$, while $\partial_3(0) = 0$. So all of $\text{im } \partial_3$ sits at nonzero momenta and the quotient is supported entirely at Γ : the Γ block is H_2 , and the 3 is $b_2(T^3)$. The two routes do not agree by accident; they are the same decomposition read in two bases.

check: the sheets average to harmonic representatives, and the Gamma block IS b.2

3.1 The wrapping sheets are cycles, not harmonic — and their harmonic representatives

Theorem 2 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x: x_m=0} [x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \text{im } \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s, \partial_3 \partial_3^\dagger s \rangle}{\|s\|^2} = 2 \quad \text{exactly, for every } L \geq 2. \quad (7)$$

check: FINDING: the wrapping sheets are cycles but NOT harmonic

The distinction matters because H_2 is used in two senses: the quotient in Theorem 2, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in.

Harmonic representatives do exist, and they are one line away. Averaging a sheet over its L translates in the complementary direction,

$$h_{(ij)} = \frac{1}{L} \sum_{r=0}^{L-1} s_{(ij)}(r), \quad \partial_2 h_{(ij)} = 0, \quad \partial_3^\dagger h_{(ij)} = 0, \quad (8)$$

gives three harmonic chains spanning $\ker \partial_2 \cap \ker \partial_3^\dagger$, which is exactly three-dimensional; and $s_{(ij)} - h_{(ij)} \in \text{im } \partial_3$, so a sheet and its average are the same homology class. The averaging is the $k = 0$ Fourier projection, and what it removes is precisely the non-harmonic part measured above. Nothing about the finding is withdrawn — the generators Theorem 2 exhibits are still not harmonic — but a real-space statement phrased for the harmonic subspace now has representatives to point

at, and Proposition 28 covers both, since $B^\dagger w = 0$ holds on all of $\ker \partial_2$.

check: the sheets average to harmonic representatives, and the Gamma block IS b.2

4 Exact all-rank dynamics at second order

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F, \quad (9)$$

with dimensions and quadratic Casimirs

$$\begin{aligned} d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0, \\ d_{\text{Adj}} &= N^2 - 1, & C_{\text{Adj}} &= N, \\ d_{\Lambda^2 F} &= \frac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \frac{(N+1)(N-2)}{N}, \\ d_{\text{Sym}^2 F} &= \frac{N(N+1)}{2}, & C_{\text{Sym}^2 F} &= \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (10)$$

Lemma 6 (The channel list is complete). *For every $N \geq 3$, a second-order process through one shared link has its intermediate shared-link representation in $\{\mathbf{1}, \text{Adj}, \Lambda^2 F, \text{Sym}^2 F\}$ up to conjugation, and in nothing else. The conjugate of a channel carries the same Casimir and hence the same weight, which is why four labels suffice where the link-resolved census of Section 5 needs six.*

Proof. The shared link of a one-plaquette shell state carries F or $\bar{F}$ by (4); one plaquette action tensors it by F or $\bar{F}$; and (9), together with the conjugate product $\bar{F} \otimes \bar{F} = \Lambda^2 \bar{F} \oplus \text{Sym}^2 \bar{F}$, decomposes every case completely and multiplicity-free. There is nothing else for the link to carry. $\square$

check: each channel gap is C.F + C.R/2, and the weights sum to one

The corollary that makes this checkable rather than merely stated is that the weights d_ρ/N^2 sum to one within each orientation family, which is arithmetic on (10): a missing channel would leave a deficit. What Lemma 6 does *not* settle — and Section 11 says so — is that one-shared-link processes exhaust the second-order off-diagonal intermediates. That is a statement about the process enumeration, not about this list.

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

check: each channel gap is C.F + C.R/2, and the weights sum to one

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 7 (Shared-link channel weights). *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2 , for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\text{Tr}(AU)\text{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-(2, 2) moment of U .

Second, the group. The Weingarten values below are the $U(N)$ ones, and this is a paper about $SU(N)$; the reduction is a step, not a slip. A degree-(2, 2) integrand is invariant under $U \mapsto zU$ for $z \in U(1)$, so integrating it over $SU(N)$ and over $U(N) = (SU(N) \times U(1))/\mathbb{Z}_N$ give the same number provided the $SU(N)$ -invariants at this degree are exhausted by the permutation contractions — which holds exactly when the degree is below the rank, $2 < N$. That is the theorem’s own range, $N \geq 3$. At $N = 2$ the extra invariant $\epsilon \otimes \epsilon$ appears and the reduction fails, which is a second reason the construction begins at $N = 3$; it is the same ϵ channel that makes the u^1 row of Appendix C $SU(3)$ -only. This step has no machine check, and Section 11 records it.

Third, two moments settle both families. In the Weingarten calculus, whose asymptotic origin is [3] and whose finite- N closed forms are [13], the order-two values are $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$ — the inverse of the S_2 Gram matrix $\begin{pmatrix} N^2 & N \\ N & N^2 \end{pmatrix}$ — the index sums give

$$\begin{aligned} M_{\text{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\ M_{\text{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \end{aligned} \quad (11)$$

For the like family, $\text{Tr}(AU)\text{Tr}(BU)$ decomposes into $\text{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\text{direct}} \pm M_{\text{cross}})/2$. Normalising,

$$n_{\text{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\text{Sym}^2 F}}{N^2}, \quad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij}\bar{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1 against $M_{\text{direct}} = N^2$, so $n_1 = 1/N^2 = d_1/N^2$ and $n_{\text{Adj}} = 1 - 1/N^2 = d_{\text{Adj}}/N^2$. $\square$

check: the shared-link weights are Weingarten, not an isotropy assumption

Remark 8. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (11) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times $N^2 - 1$ is 1 identically in N , so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these are why the all-rank second-order result is self-contained rather than conditional. Appendix B records the index sums.

check: the Weingarten route is independent of the corpus

Remark 9 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [15] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim \rho / (\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N . That is provenance for a chain already checked here, not a second derivation, and it does not check the premise that four channels exhaust the shared-link routes.

check: the published dimension-ratio matrix element is this registry’s weight formula

With Theorem 7 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \quad (12)$$

and summing within the two orientation families,

$$A_N = w_1 + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad (13)$$

$$B_N = w_{\Lambda^2} + w_{\text{Sym}^2} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (14)$$

Each individual weight in closed form is Appendix A.

check: the four channel weights follow from dimension and Casimir

check: A.N and B.N are the channel sums, not transcriptions

4.2 The shared-link law

The statement that follows is about *one matrix element*. Promoting it to the whole order- u^2 operator on the torus is a separate step with its own hypotheses, and the two are separated here because the paper’s own Scope section lists one of those hypotheses as unchecked.

Theorem 10 (All-rank adjacent shared-link coefficient). *For every integer $N \geq 3$, in the untruncated*

theory or in any regulator retaining all four fusion channels of (9), the connected second-order matrix element between two plaquettes sharing one oriented link is $t_N u^2$ times the signed incidence overlap of that pair, with

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0.$$

Proof. Every second-order process between two plaquettes sharing exactly one link runs through that link's fusion channel; the list is complete by Lemma 6, the weight of each is Theorem 7, and the resolvent denominator is $C_F + C_\rho/2$. Summing within the two orientation families gives (13) and (14), and the relative orientation of the pair enters as the incidence product, so the two families appear with opposite sign and the element is $(B_N - A_N)u^2$ times that product. Positivity: for $N \geq 3$ every denominator factor is positive and $N^2 - 4 > 0$. The regulator premise is not decoration — Corollary 16 exhibits a published cutoff under which the same element comes out $-1/12$, of the opposite sign. $\square$

check: $t_N = B_N - A_N$

check: $t_N > 0$ for $N \geq 3$

Conditional corollary 11 (Incidence assembly on the finite torus). Assume in addition that (i) the range of P_- is exactly the charge-odd one-plaquette span, and (ii) after the linked, vacuum-subtracted fold of (5) no connected second-order history between *non-adjacent* plaquettes contributes a momentum-dependent term on P_- . Then

$$H_{\text{eff},-}^{(N)}(k, u) = E_{\text{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3),$$

with $E_{\text{flat},N}$ momentum independent, and the three bands through second order are $E_{\text{flat},N}(u)$ and $E_{\text{flat},N}(u) + t_N u^2 q(k)$ twice.

Proof. Under (ii) only shared-link pairs carry momentum dependence, and Theorem 10 gives each of those elements, so the off-diagonal part is $t_N S$ with $S = BB^\dagger - 4I$. The diagonal is a scalar, and for a reason worth naming because it fails elsewhere in this paper: translations act transitively on the faces of each orientation and the cubic point group permutes the three orientations, so the faces are a *single* orbit and any invariant diagonal is a multiple of the identity. On an open cluster that is false, and Proposition 18 is what replaces it there. Absorbing $-4t_N$ into the scalar and applying Theorem 1 gives the bands. $\square$

check: $B B^\dagger = q I - d \text{ conj}(d)^T$ for the curl incidence

Hypothesis (i) is the retained-shell premise of Section 2; (ii) is the one Proposition 12 narrows but does not close. Neither has a check, and Section 11 says so.

Two consequences,

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \dots \quad (15)$$

check: $t_2 = 0$ and $t_3 = 5/612$

check: large- N expansion of t_N through $1/N^9$

4.3 The vacuum-mediated route

Hypothesis (ii) of Corollary 11 is not vacuous, and one family of routes it has to exclude can be excluded outright. The vacuum-mediated route connects *any* pair of plaquettes, adjacent or not, and it vanishes in this sector by a charge-conjugation selection rule. That closes one family and not the hypothesis; omitting it is a recorded error, not a hypothetical one.

Proposition 12 (Vacuum-mediated route). *The route $\langle p'|V|0\rangle\langle 0|V|p\rangle/E_0$ is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p,-\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such term, while the charge-even shared-link element acquires $1/C_F$:*

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2 - 5)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

check: $\ell_N = A_N + B_N + 1/C_F$, the vacuum-mediated route at every rank

Remark 13. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

One thing this proposition does *not* do is dispose of the route between non-adjacent pairs. Taken over all pairs it would be a rank-one operator supported at $k = 0$, which is not the twelve-neighbour adjacency ℓ_N multiplies; which contributions survive in H_2 is settled by the linked, vacuum-subtracted prescription of (5), not by anything derived here. What the check above establishes is the all-rank value ℓ_N of the shared-link element, and the disposition of the non-adjacent contributions is prescription rather than result.

4.4 Finite-volume width

Let $q_{\text{max}}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\text{max}}(L) = \begin{cases} 12, & L \text{ even,} \\ 12 \cos^2(\pi/2L), & L \text{ odd,} \end{cases} \quad (16)$$

because only at even L does some k_j reach π exactly. Under the hypotheses of Corollary 11, the width of the retained charge-odd manifold is $W_{N,L}^{(-)} = q_{\max}(L) t_N u^2 + O(u^3)$, so for even L and in the thermodynamic limit $[u^2]W^{(-)} = 12t_N \sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

check: the zone maximum of q is 12 only at even L

The smallest nonzero incidence eigenvalue is $q_{\min} = 4\sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

check: $q_{\min}$ on the L -torus grid is $4 \sin^2(\pi/L)$

4.5 The orientation signs are essential

Removing the orientation signs — from the entries of (6) and from the phases, so each $e^{ik} - 1$ becomes $e^{ik} + 1$ — gives an unsigned incidence $\mathcal{N}(k)$ with $A + 4I = \mathcal{N}^\dagger$. The contrast is a determinant identity, not a sample: read straight off the boundary formulas, $\det B(k) \equiv 0$ identically in k — a zero eigenvalue at every momentum, which is the whole flat band — while $\det \mathcal{N}(k) = -2 \prod_j (1 + e^{ik_j})$ vanishes only where some $k_j = \pi$. (Sampling agrees: the unsigned adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue.) The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

check: the two Bloch incidence symbols, and the determinant asymmetry between them

This control is not hypothetical: the unsigned spectrum is the charge-even sector's, which Section 6 computes in closed form, and the two spans are the two bandwidths. The signed span is $8 - (-4) = 12$, giving the charge-odd width $12t_- = 5/51$ at $N = 3$; the unsigned span is $12 - (-4) = 16$, giving the charge-even width $16|\ell_3| = 88/153$. Both spans carry the even- L hypothesis of (16), and for the same reason: the charge-odd 12 needs every $k_j = \pi$ and the charge-even floor -4 needs some $k_j = \pi$, which the periodic grid samples only when L is even. The charge-even *top* needs no hypothesis — it sits at Γ . Every bandwidth, ratio and band edge quoted below is therefore an even- L , thermodynamic-limit statement unless it is a Γ statement.

check: the C-even floor needs even L too, so both bandwidths are even- L statements

check: the two band spans ARE the two incidence spectra

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That

ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3, 2, 2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $B = 6$ $SU(3)$ truncation, of dimension 1,590,462. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven — a census result and not the plaquette count. The full E_* eigenspace has dimension 22, split $11_{C=+} \oplus 11_{C=-}$ by tensor-derived charge conjugation, and the delivery removes all 22 from the pseudoinverse; because the eleven columns exhaust the odd half, that agrees with the $Q = 1_- - P_-$ prescription of (5). Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\text{conn}} = K_{LR} - J_L K_L J_L^\dagger - J_R K_R J_R^\dagger + J_F K_F J_F^\dagger, \quad (17)$$

and the same fold applied to the geometry gives G_{conn} . The fold is literal in the sense that no eigenvalue or fitted scalar is subtracted: each source operator is folded in its own coordinates and lifted. The delivery corroborates that internally by applying the same weights history by history over its 1,984 ordered two-step histories, of which 1,192 cancel at weight zero, and the two routes agree to 4.9×10^{-15} .

First order settles itself. On the shell $PVP = I_{11}$, so the first-order term is nonzero but scalar: it shifts all eleven branches equally and creates no branch ambiguity. Its connected fold then vanishes *identically*, and by counting rather than by cancellation — each of the eleven faces is covered exactly once by $\{L, R\}$ once the shared face is restored by F , so every Möbius weight in (17) applied to the identity is $1 - 1 = 0$. Second order is therefore the leading connected term and not a correction to a first-order one.

check: connected first order vanishes by inclusion-exclusion, not by cancellation of numbers

At second order the result is

$$K_{\text{conn}}^{(2)} = \frac{5}{612} G_{\text{conn}} + D, \quad (18)$$

with D diagonal and delivered. The geometry alone fixes the *splitting* inside each block and nothing else: G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), so each pair splits by $2 \times 5/612 = 5/306$ about its own diagonal entry. Given D , which Appendix F prints, the spectrum is then closed form, $\{0, (-\frac{15}{4})^2, (-\frac{34}{9})^4, (-\frac{129}{34})^4\}$ — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind —

supplied to no builder, graph fit, branching rule or validator — and reports that the payload hashes were produced before the recorded comparison, a chronology it itself calls record-backed and only partly auditable; the value is byte-present in the nested $B = 4$ package that the $B = 6$ build seals as an authority object, so the defensible statement is dependency-path nonuse and not corpus-wide absence. This repository and this repository re-thresholds those gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects: on a 66-dimensional held-out word space the correct second-order residual falls with log-log slope 2.90 while the sign-reversed control falls with slope 2.00, a minimum error ratio of 325. What all this establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration.

check: the connected two-cube geometry has exactly four cross-cell pairs, each -1

check: the B=6 connected kernel is (5/612) $G_{\text{conn}} + \text{diag}$, with the certified spectrum

check: the certificate's own gates pass, target-blind, with the wrong-sign control rejected

Resolving (18) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

$$\begin{array}{c|cccccc} \rho & \mathbf{1} & \mathbf{3} & \bar{\mathbf{3}} & \mathbf{6} & \bar{\mathbf{6}} & \mathbf{8} \\ \hline c_\rho & \frac{1}{12} & -\frac{1}{12} & -\frac{1}{12} & -\frac{1}{9} & -\frac{1}{9} & \frac{16}{51} \end{array} \quad (19)$$

check: the B=6 six-channel census sums to the registry's own $t_{.3} = 5/612$

That these six sum to $5/612$ is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (19) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.2×10^{-15} per channel — 2.1×10^{-14} across the delivery's reconstructed matrices as a whole — and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 7.

Reported computational result 14 (The census is the Weingarten ledger). At $N = 3$ the six reconstructed link-irrep coefficients of (19) are the four fusion-channel weights (12) individually:

$$\begin{aligned} c_{\mathbf{1}} &= -w_{\mathbf{1}}, & c_{\mathbf{8}} &= -w_{\text{Adj}}, \\ c_{\mathbf{3}} &= c_{\bar{\mathbf{3}}} = \frac{1}{2}w_{\Lambda^2 F}, & c_{\mathbf{6}} &= c_{\bar{\mathbf{6}}} = \frac{1}{2}w_{\text{Sym}^2 F}. \end{aligned}$$

Substituting (10) into (12) at $N = 3$ gives $w_{\mathbf{1}} = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\text{Sym}^2 F} = -2/9$; comparison with (19) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

check: the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel

One further feature of the census is worth separating out, because it is the paper's central architecture measured rather than assumed. The six link-resolved matrices are not merely six numbers that sum correctly: each is *separately* proportional to the same incidence Gram. The eleven oriented plaquettes of the prism have exactly 56 ordered adjacent pairs, each sharing one link with signed overlap ± 1 ; the delivery reports a constant channel-to-geometry ratio on all 56 of them, for every one of the six channels, with worst residual 2.2×10^{-15} . Colour dynamics fixes a number per channel and the cellular boundary operator fixes the matrix; if the two mixed, the ratio would be constant for the total and not for the parts.

check: every shared-link channel is separately proportional to the geometry, on all 56 pairs

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2 . Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $B = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

It is scoped in one further way, which matters more. The two-cube build draws its local Wilson amplitudes from the same hash-pinned upstream Clebsch–Gordan archive as the single-cube reconstructions of the next subsection, and the squared amplitudes that archive produces are the published d_ρ/N^2 [15] — the numerator of (12). The census therefore does not confirm the weights from an independent source: it tests the *assembly* that carries them, which is the reachable-channel list, the resolvent denominators, the operator-level fold, the even split between conjugates, and the geometry that survives each channel separately. That is the statement worth making, and it is a different statement from confirmation.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate — the two orientations of the shared link weighted equally — which the delivery resolves on the two-cube space rather than assuming. It is not measured *here*: what this repository checks is the identity the six delivered

numbers satisfy.

Conjecture 15 (All-rank even split). For every $N \geq 3$ each like-family fusion weight splits evenly between an irrep and its conjugate in the link-resolved census.

This is a conjecture and nothing here promotes it. Falsifying it means a link-resolved census at some $N \geq 4$ in a truncation that still retains every adjacent shared-link channel; at $N = 4$ the endpoint budgets make that $B \geq 33/4$, and no such construction has been run.

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3 . An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation, meaning $C_2(\rho) + 2C_2(\mathbf{3}) \leq B$: the sextet needs exactly 6 and the adjoint $17/3$, so both survive $B = 6$ and neither survives $B = 4$. The weak inequality is not a convenience. Both retentions this section uses are decided by an *equality*: $\Lambda^2 F = \mathbf{3}$ has budget exactly 4 and so sits exactly on the $B = 4$ cutoff, and the sextet sits exactly on the $B = 6$ one. Under a strict rule $B = 4$ would reach $\{\mathbf{1}\}$ alone and $B = 6$ would lose the sextet, contradicting both delivered censuses — so the deliveries are what fix the convention, which is also why the budget arithmetic is not an independent confirmation of either census.

check: B=6 retains every adjacent shared-link channel;
B=4 provably cannot

check: the retention rule is $C_2(\rho) + 2 C_2(\mathbf{3}) \leq B$, and both retentions it decides are equalities

Corollary 16 (What a link cutoff drops). *Retaining only the singlet and $\Lambda^2 F$ routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_{\mathbf{1}} = -\frac{1}{12},$$

opposite in sign to t_3 and 10.2 times larger in magnitude. What is dropped is

$$w_{\text{Sym}^2 F} - w_{\text{Adj}} = \frac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$. This is arithmetic on (12) alone.

The published $p+q \leq 1$ link cutoff retains exactly those two routes — a per-link restriction, where $B = 4$ is a total endpoint budget, so the two schemes agree on the retained channels at this order and are not the same rule — so $-1/12$ is its projected hopping and $14/153$ is the counterterm that completes it — not $5/612$, which would double-count what it already carries. On the two-cube space the first of those two numbers is delivered rather than inferred: a separately sealed $B = 4$ run

on the identical geometry gives $K_{\text{conn}}^{(2),B4} = -\frac{1}{12}G_{\text{conn}} + D_{B4}$ outright, and the Casimir budgets of the previous paragraph say without reference to any census that $B = 4$ can reach only $\{\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}\}$. Conditional on (19), the same split of the census reproduces it, $c_1 + c_3 + c_{\bar{\mathbf{3}}} = -1/12$, and gives the omitted half as $c_6 + c_{\bar{6}} + c_8 = 14/153$.

check: the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel

check: FINDING: the T1 link cutoff reverses the sign of t_3 , and 14/153 is what it omits

Remark 17. Two single-cube reconstructions bracket the same statement. A full $B = 4$ open cube assembled from the published `ymcirc` $d = 3$, $B = 4$ plaquette table [16] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11} ; a $B = 6$ cube reproduces $+5/612$ and the inverse ordering to 8.4×10^{-15} . Both locate the missing weight at the same $14/153$.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage — and the two-cube build draws its Clebsch–Gordan amplitudes from the same hash-pinned `pyclebsch` archive as the $B = 6$ cube, so that agreement is a cross-check of assembly and not of the amplitudes. The published closed form [15] is d_ρ/N^2 — the very numerator of (12). This is a cross-check of assembly, not a third confirmation of t_3 ; the exact rational identity is Corollary 16, which needs no tolerance. The $p+q \leq 1$ nomenclature and the finite-rank matrix element are both from [15], and single-cube $SU(N)$ diagonalisation in this style goes back to [12].

A counterterm added to a published T_1 Hamiltonian is therefore $14/153$ and not $5/612$, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight $w_{\mathbf{1}}$, and the cutoff value is a *difference* of two weights, not a weight.

check: FINDING: a full $T_1 = B = 4$ cube Hamiltonian reproduces $-1/12$ and the reversed shell

check: FINDING: the $B = 6$ cube flips the sign back to $+5/612$, closing the decisive test

The $B = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is $39/68$ against the bridge’s $11/34$, differing by exactly $1/4$, the local same-face sextet route, which first enters at cutoff $20/3$ — integer label $B = 7$, so any cutoff above $20/3$ already carries it. That is not only arithmetic: the delivery reports the prediction executed. A six-face $B = 7$ reachable-state probe returns $\alpha_{B=7} = 11/34$ and $t_{B=7} = 5/612$ with scalar-plus-Gram residual 1.1×10^{-15} , so raising the cutoff past the same-face threshold lowers all three absolute coefficients by exactly $1/4$ and leaves the relative shell $\{0, 5/153, 5/102\}$ untouched — the scalar moves, the hopping does not. That probe’s artifact did not travel, so it is reported here and not checked.

check: the $B = 6$ scalar misses the bridge's by exactly the same-face sextet route

5.2 What the cutoff moves, and where

The connected diagonal D is eleven numbers in both deliveries and is treated as a remainder in both. It is not eleven numbers. The open prism has a symmetry group — the cell exchange $x \mapsto 2 - x$, the two transverse reflections and the $y \leftrightarrow z$ swap, of order 16 — and D is constant on its orbits, which are $8 + 2 + 1$: the eight side faces, the two end caps, the shared face.

Proposition 18 (Where the truncation lives). *The eight-element orbit is exactly the support of the four cross-cell pairs, and it is the only orbit whose diagonal value differs between the two truncations: the end caps carry $-15/4$ and the shared face 0 at both $B = 4$ and $B = 6$, while the side value moves from $-7/4$ to $-2317/612$. Together with the four cross-cell blocks carrying the whole off-diagonal change, the entire $B = 4 \rightarrow B = 6$ difference is confined to the faces that carry connected transport.*

check: the connected diagonal is orbit-constant, and only the transporting orbit moves

This also says what the nonuniform D is. Where the faces of a complex form a *single* orbit — the closed cube of the next subsection, the periodic torus of Theorem 10 — orbit-constancy forces the diagonal to be scalar, and the operator is exactly $\alpha I + tG$. The prism has three orbits because it has a boundary. D is the open boundary showing up in the diagonal, not a failure of the incidence form.

5.3 The one-cube shell, and why its flat level cannot move

Those two single-cube reconstructions are usually read as numerical cross-checks. They are more than that, because the object they diagonalise is the same chain complex. The six faces of a cube are a closed surface, so in the outward orientation $\ker \partial_2$ is one-dimensional and spanned by the sum of all six — the fundamental class, $b_2(S^2) = 1$. That is Theorem 2 on a complex with no three-cells, where the $L^3 - 1$ boundaries are absent and only the class survives.

Proposition 19 (One-cube shell). *On the closed cube surface $G = \partial_2^\top \partial_2$ commutes with all 24 proper rotations. The combinations $b_{+i} - b_{-i}$ carry the defining vector representation and the combinations $b_{+i} + b_{-i}$ the permutation representation of the three unordered axes, so with the charge-odd inversion $Pb_{+i} = -b_{-i}$ the shell is*

$$A_1^{--} \oplus T_1^{+-} \oplus E^{--},$$

with G -eigenvalues 0, 4, 6 and multiplicities 1, 3, 2. Consequently $\alpha I + tG$ has levels α , $\alpha + 4t$, $\alpha + 6t$.

Proof. The rotations permute the six outward faces, and G is built from the boundary map, so it commutes with the permutation action. On the antisymmetric combinations that action is the defining representation by inspection — $b_{+i} - b_{-i}$ transforms as e_i — and on the symmetric combinations it is the permutation representation of $\{\pm e_i\}$, which is $A_1 \oplus E$. Under $Pb_{+i} = -b_{-i}$ the antisymmetric sector is parity even and the symmetric one parity odd. In these sectors G is $4I$ and $4I - 2(J - I)$ respectively, of spectra $\{4, 4, 4\}$ and $\{0, 6, 6\}$; since the three isotypic components have distinct dimensions the eigenvalues attach to them uniquely. $\square$

check: the one-cube shell is $A_1 + T_1 + E$, and its flat level is the S^2 fundamental class

Three things follow. The cubic labels are derived here rather than assigned: both deliveries assert them and neither builds the symmetry matrices. The A_1^- level is Proposition 28's mechanism on a second complex — $Gw = 0$ there, so that level sits at α for every t , and no truncation, no channel and no coefficient in dispute anywhere in this paper can move it. And the reversal the two runs report is therefore a statement about the sign of one number rather than about the shell as a whole: at $B = 6$ the levels are $\{39/68, 371/612, 127/204\}$ and at $B = 4$ $\{37/12, 11/4, 31/12\}$, in the opposite order, with A_1^- in each case sitting exactly at the scalar. The relative shell $\{0, 4t, 6t\}$ is $\{0, 5/153, 5/102\}$ at $B = 6$.

The finite-volume fingerprint of the 28 August finite- N bridge certificate — a different release, and named here because a reader tracing it to the two-cube one would find nothing — is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence spectrum is $\{0^{29}, 3^{12}, 6^{24}, 9^{16}\}$, and $29 = L^3 + 2$ is Theorem 2.

check: the certificate's finite-volume fingerprints, and $29 = L^3 + 2$ is the Lean cycle count

Remark 20 (Scope). Everything in this section is second order in u , finite volume, strong coupling, one sector. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The $B = 6$ two-cube construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B -truncated and not proved cutoff-stable. The held-out finite- u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space with the $Q_1 - Q_1$ magnetic block set to zero, not the full Hamiltonian, and no radius-three stability claim was executed. The two branches are not equally covered: at $B = 4$ the delivery reports a direct diagonalisation of the complete 4,180-dimensional charge-odd block at four small

couplings, two of them held out, with the residual after $E_* + u + u^2 \text{ eig } K_2$ scaling as u^3 ; at $B = 6$ no such diagonalisation exists and the 66-dimensional star stands in for it. That $B = 4$ report carries no check here — its certificate did not travel — and is recorded as the delivery's rather than as this paper's. No external group has reproduced the release.

6 The charge-even sector in closed form

The unsigned incidence $\mathcal{N}(k)$ of Section 4 is not a hypothetical control: it is the charge-even sector's Bloch symbol. Write

$$\hat{a}_m = 2 + 2 \cos k_m = |1 + e^{ik_m}|^2, \quad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4 \sin^2(k_m/2)$ used in the charge-odd shape basis (21) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 21 (Charge-even Bloch cubic). $p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{N}\mathcal{N}^\dagger$ is

$$\mu^3 - 2p\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the charge-even adjacency eigenvalues $\lambda = \mu - 4$. The charge-odd counterpart of the same computation is $\mu(\mu - q)^2$.

check: the C-even characteristic polynomial is $\text{mu}(\text{mu} - p)^2 = 4 \text{ a}_1 \text{ a}_2 \text{ a}_3$

check: $p + q = 12$: one zone function runs both sectors

The closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

check: the Bloch cubic IS the finite $L = 3$ and $L = 4$ plaquette spectrum, exactly

Proposition 22 (Range and attainment). $\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \leq 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $f(16) = 16(16 - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3 \geq 0$ because $p \leq 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 \leq 64$. Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ . Equality at the bottom

is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. $\square$

check: the C-even range $[-4, 12]$ is exact; the top only at Γ , the floor on three planes

check: the C-even band touches its floor exactly on the three planes $k_j = \pi$

Remark 23. A flat charge-even branch would need $f(\mu_0) = 0$ identically in k , which the constant term already forbids. The charge-even kernel is three measure-zero planes against the charge-odd kernel's whole band: the two sectors are two readings of one geometric object, and only the signed one is flat.

check: the C-even band touches its floor exactly on the three planes $k_j = \pi$

check: the C-even spectra at the four high-symmetry momenta

6.1 One assembly, both sectors, both orders

The $r = 2$ content below is all-rank and checked. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

The scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\text{tower}_{r,s}$ for the within-plaquette term in canonical u ,

$$E_s(\lambda, r) = \text{tower}_{r,s} + 12 \text{leak}_{r,s} + \lambda t_{r,s}, \quad (20)$$

with $\lambda \in \{-4, 8\}$ in the charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the charge-even sector, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta(3u/2)$, so (20) takes no unregistered input. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link. The two towers are different objects and both are needed: the charge-odd one contributes $8/3 + u + u^2/2 + 7u^3/32$ and the charge-even one $8/3 - u + (13/20)u^2 + (101/200)u^3$, each of them $4\Delta(3u/2)$ on its own sector's series, so $\text{tower}_{2,-} = 1/2$ against $\text{tower}_{2,+} = 13/20$ and $\text{tower}_{3,-} = 7/32$ against $\text{tower}_{3,+} = 101/200$. The nine values it reproduces are tabulated in Appendices C and D.

check: one assembly formula gives every registered band value

check: one assembly formula gives every C-even value at both orders

check: the plaquette graph is 12-regular and two faces share at most one link

Because $\text{leak}_{r,+} = t_{r,+}$ at both computed orders, the whole charge-even sector collapses to $\text{tower}_{r,+} + (12 + \lambda) t_{r,+}$, a one-parameter family in the charge-even tower. That equality is verified and unexplained. The order-three separation is between the charge-odd leakage and the charge-even hopping — not, as an earlier edition printed, between the charge-odd leakage and the charge-odd hopping, which were never equal: $\text{leak}_{2,-} = t_{2,+} = -11/306$ against $t_{2,-} = 5/612$ at second order, and $\text{leak}_{3,-} = -12331/249696$ against $t_{3,+} = -6335/249696$ at third. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

check: both declared coincidences, checked: one is ell_N at all ranks, the other is bare

6.2 Where the rest frame is not blind

Proposition 24 (Charge-even Γ splitting). *At Γ the charge-even spectrum is $\{12, 0, 0\}$, so the A_1^{++} level and the E^{++} doublet are distinct, and*

$$E(A_1^{++}) - E(E^{++}) = 12 t_{r,+}$$

exactly: $-22/51$ at $r = 2$ and $-6335/20808$ at $r = 3$.

check: the C-even Gamma point pins $t_{r,+}$, exactly where no C-odd Gamma datum can

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned spectrum has two levels, so the charge-even rest frame can. Neither half needs the incidence ansatz, and stating them through it understates both. At Γ the little group is the full cubic point group, so the 3×3 block commutes with its action at every order in u and whether or not the correction factors through links; Schur then decides the two sectors, and decides them differently. The charge-odd state is the *oriented* 2-cell $e_i \wedge e_j$, on which a proper rotation acts by its exterior square, which is the rotation itself under Hodge duality — the defining rep T_1 , irreducible, commutant dimension one, so the block is a scalar. Charge conjugation drops the orientation, so the charge-even state is the *unoriented* plane and the action is the permutation of the three axes, $A_1 \oplus E$, commutant dimension two. The two commutant dimensions are exactly the two level counts. The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — $22/459$ at second order and $6335/187272$ at third.

check: the C-even curvature is $(4/3)|t_{r,+}|$ at both orders, and isotropic

check: Schur alone fixes both Gamma blocks: the C-odd triple is irreducible, the C-even one is not

At both computed orders the charge-even bandwidth is $16|t_{r,+}|$ and the charge-odd manifold width $12|t_{r,-}|$. At any order at which the correction stays in the incidence algebra — which Proposition 28 warns is not automatic — the second-order ratio is $4(3N^2 - 5)/[3(N^2 - 4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from $88/15$ at $N = 3$ towards 4 — a ratio of two even- L spans, in the sense fixed above. So at second order the charge-even band is the wider one at every rank, and that asymmetry is a channel statement, not a geometric one.

check: the C-even bandwidth is $16|t_{r,+}|$ at every order; the C-odd manifold width is $12|t_{r,-}|$

check: at $N = 2$ the C-odd hopping vanishes and the C-even one does not

7 $SU(3)$ through third order

The *third-order* content of this section is conditional on supplied exact-arithmetic ledgers. Its second-order content is not: leak_2 is the all-rank ℓ_N of Proposition 12 at $N = 3$, and the tower is the certified coupling conversion. The within-plaquette charge-odd tower contributes $8/3 + u + u^2/2 + 7u^3/32$, and the registered $SU(3)$ values, with the scope of each row, are collected in Appendix C.

Reported computational result 25 (Third-order ledger). The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

check: $d_3 = 7/32 + 12*\text{leak}_3 - 4*b_3$

check: leak_3 is assembled from the domino diagonal and the vacuum piece

Conditional corollary 26 (Third-order factorisation). If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\text{eff},-}(k, u) = E_{\text{flat}}(u)I + \tau(u)B(k)B(k)^\dagger + O(u^4),$$

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \\ \tau(u) = \frac{5}{612}u^2 + \frac{1975}{124848}u^3.$$

check: E_{flat} and $t(u)$ carry the ledger coefficients

The second-order diagonal is the $\lambda = 0$ value $\frac{1}{2} + 12\text{leak}_2 = 7/102$ with $\text{leak}_2 = -11/306$; subtracting $4t_-$ gives the flat scalar $11/306$ at $\lambda = -4$.

check: $d_- = 1/2 + 12*\text{leak}_2$, and $\text{leak}_2 = -11/306$

Remark 27 (What the external comparison can and cannot do). Hamer’s 1989 rest-frame series [11] agrees with E_{flat} at every shared order. It cannot test τ . The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ -point datum is blind to it. That argument runs inside the incidence ansatz, which Proposition 28 warns is not unconditional; the Schur argument of Section 6 removes the caveat, since the charge-odd triple carries the irreducible T_1 and its Γ block is therefore scalar at every order. The charge-even rest frame is not blind, by Proposition 24; the one-parameter family $(b_3 + \varepsilon, \text{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\text{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (20). Proposition 24 is the charge-*even* analogue and separates leak_+ from t_+ ; it says nothing about b_3 or leak_3 .

check: FINDING: no Gamma-point datum can constrain the hopping

8 What the homology protects

Proposition 28 (Boundary-factorised rigidity). *If an order- r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 1. The middle term is annihilated whatever M_r is, which is the content: the protection belongs to the boundary operator and not to the dynamics. $\square$

check: a boundary-factorised correction shifts the carrier without dispersing it, for generic M

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order correction need not factor through links. The carrier-projected fourth-order shapes the corpus works with are

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \quad (21)$$

with $e_2 = \sum_{i < j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4 \sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 1, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger H_4 w)/q$ and the numerator, not ε_4 , is the polynomial.

check: the carrier projection is where the $1/q$ comes from

check: clearing the denominator reproduces the five-element numerator basis

That family is *not* what cubic symmetry permits, and an earlier edition of this paper attributed it there. The point group permutes (a_1, a_2, a_3) , so its invariants are the symmetric polynomials, and the numerators of degree at most three with no constant term span a six-dimensional space $\{q, q^2, e_2, q^3, qe_2, e_3\}$ — one per partition. Clearing the denominator in (21) returns five of those six; the missing direction is q^3 , that is the shape q^2 , which is cubic invariant and regular at Γ ($q^2 \sim |k|^4$), so neither the point group nor the carrier projection excludes it. The omission is a property of the two-hop enumeration that produced the family, and it is load-bearing: Theorem 29 is a rank statement about *this* span, and a sixth direction would be a sixth unknown for the same data to determine.

check: the shape family is not symmetry-complete: cubic symmetry alone permits q^2 as well

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient C_{shp} . This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What it does is state precisely where the disagreement lives, because that geography is now checked and it is what a decisive calculation must target.

The two values, side by side and in no order of preference:

$$\begin{aligned} C_{\text{old}} &= -\frac{211835444920651}{4405310420659200} = -0.0480863\dots, \\ C_{\text{new}} &= -0.020213328886166577, \\ \Delta_C &= C_{\text{new}} - C_{\text{old}} = 0.0278730\dots \end{aligned} \quad (22)$$

The first is recorded as an exact rational and the second only as a float — that asymmetry is a fact about the two records, not an argument for either. The gap is what “less than half the disputed gap” and “the two records give $K = 17.04$ and $K = 23.66$ ” below are measured against.

check: $\Delta_C = C_{\text{new}} - C_{\text{old}} > 0$ (the real discrepancy)

The two records do agree on the axial datum, but the corpus’s own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13} . The sealed core holds internally — α_{new} and $4A_{\text{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

check: v10a.26 A, B, D match the sealed rationals within $2.3\text{e-}13$

check: FINDING: alpha_new falls outside the corpus's own 2.3e-13 bound

9.1 The obstruction is polynomial

Theorem 29 (Obstruction certificate). *The two recorded off-axis coefficients enter (21) only through the shape $4e_2/q$, so the two records' carrier numerators differ by $4\Delta_C e_2$ and their bands by $\Delta_C (4e_2/q)$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the zero polynomial there and both differences vanish identically. Consequently no Γ -point or axial datum distinguishes the two records, at any precision. Off axis they do differ: $e_2(M) = 16$ and $e_2(R) = 48$, so the numerators differ by $64\Delta_C$ and $192\Delta_C$ and the bands by $8\Delta_C$ and $16\Delta_C$.*

check: FINDING: the retained Gamma/axis data cannot identify C_shp

check: Phi_C vanishes at Gamma along every direction

Remark 30. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

check: the crosswalk is exactly scalar on the momentum axes

On the axial cut itself the invariants collapse to $q = L(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$, and dividing the raw cube-boundary numerator $(\alpha_N/4)L^2$ by $\|w\|^2 = q = L$ leaves a single power of L with coefficient $\alpha_3/4 = 5/48$.

check: on an axial cut the mixed invariants vanish and the norm divides

9.2 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The exhibition adds no logical strength: Theorem 29 already proves that *every* value of C is consistent with the retained data, so a witness is one point of a line already known to be a line. It is a concrete handle, not evidence. The witness below is not itself a candidate for the physical coefficient — it is the balanced ϵ -free continuation of the all-rank shape family, and ϵ -free is precisely what the physical value is not. Its job is to make the $4\Delta_C e_2$ family concrete with an exactly known shift.

Proposition 31 (Explicit second witness). *There is a value C_{alt} with $C_{\text{alt}} - C_{\text{old}} = 25/1024$ exactly, appearing verbatim in the pinned source edition, whose induced band separations are 0 at X , $25/128$ at M and $25/64$ at*

R . It is identical to C_{old} on every retained datum and distinct from it off-axis.

check: FINDING: an explicit second witness C_alt exhibits the C2 non-identifiability

check: the C_alt witness IS the balanced eps-free continuation, and Delta_C/(A/2) = 15/32

9.3 Where the coefficient lives

The four checkpoint deltas are the Γ -subtracted carrier corrections $\Delta_s = \varepsilon_4(k_s) - \varepsilon_4(0)$ at $k_X = (\pi, 0, 0)$, $k_M = (\pi, \pi, 0)$, $k_P = (\pi, \pi/2, 0)$ and $k_R = (\pi, \pi, \pi)$ — P is the off-axis, off-corner point that makes the system invertible, with $a = (4, 2, 0)$. In the shape basis they are

$$\begin{aligned}\Delta_X &= 4A, \\ \Delta_M &= 8A + 16B + 8C, \\ \Delta_P &= 6A + 8B + \frac{16}{3}C, \\ \Delta_R &= 12A + 48B + 16C + \frac{16}{3}D,\end{aligned}\tag{23}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

check: checkpoint values at X, M, P, R

check: the four extraction formulas invert the ansatz

check: X is blind to B, C, D | it fixes A alone

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15} , as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q_{\text{band}}^{(4)}$ and $m_{\Gamma}^{(4)}$ are differently anchored coordinates and not rival estimates.
2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.
3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h , and it sets both shapes rigidly: $A_{\text{normal}} = -h$

while the raw normal amplitude is $2h$, which in the shape normalisation of (21) is $C_{\text{normal}} = h/2$, so

$$C_{\text{normal}} = -\frac{1}{2}A_{\text{normal}} \quad \text{exactly.}$$

(The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A -carrying sector; no redistribution reaches the gap.

4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

check: the shape fit's C row sums to zero, so no Gamma-anchor error can move C.shp at all

check: a translation-local scalar shift changes nothing observable

check: FINDING: C.shp is carried by 6 of 189 records, not spread across the kernel

check: FINDING: $A = 5/48$ pins the normal sector's whole C4 contribution

check: $C_{\text{normal}} = -A_{\text{normal}}/2$: the agreed axial coefficient pins the normal channel

check: the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A -carrying sector rather than by redistribution. That is a constraint on candidates built on the recorded 189-record support — which is where the measurement was made; a kernel on a different support is not constrained by it at all. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.4 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ -point scalar and emits no kernel block at all, and by Theorem 29 Γ -point data are structurally incapable of constraining

C_{shp} . This is a statement about the instrument, made by reading it, not a report of a failed run.

check: FINDING: the marked-cluster engine emits the Gamma scalar only | a completed 609-sweep cannot decide C.shp

9.5 The tier collapse, as two integers

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a -degree three, because the projection contributes one further power of a : for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\text{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3 . An earlier degree-bound mechanism proposed for the collapse was therefore retracted. What is checked is finer: every degree-three block amplitude is an integer multiple of one raw record weight x , and

$$B = 0 : (+1, +1, -2) \cdot x, \quad D = 0 : (-3, +6, -3) \cdot x$$

over the displacement shells. The kernel does populate the forbidden tier; the dynamics cancels it with small integers. Any mechanism must reproduce those two vectors.

check: RETRACTED: the vertex count does NOT forbid B.shp and D.shp

check: FINDING: the tier collapse is two integer cancellations at record level

check: the vanishing coefficients are exactly the degree-3 ones

9.6 A near- Γ statement that does not adjudicate

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. What the criterion needs is a lower bound on q over the region $|k| \geq r$, and there it is sharp: writing $q = \sum_m h(k_m^2)$ with $h(t) = 2(1 - \cos \sqrt{t})$, the identity $(s \cos s - \sin s)' = -s \sin s \leq 0$ makes h concave and increasing on $[0, \pi^2]$, so the minimum over the simplex sits at a vertex and

$$\min_{|k| \geq r} q(k) = 4 \sin^2(r/2), \quad (24)$$

attained on a coordinate *axis*. With $\tau(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are therefore dominated by the second-order dispersion outside a ball of radius $2 \arcsin(\frac{u}{2} \sqrt{W_4/(\theta t_3)})$, which is Ku to leading order with $K = \sqrt{W_4/(\theta t_3)}$: the two records give $K = 10.85$ and $K = 15.06$ — floating point, since one

of the two bandwidths exists only as a float — and the larger bounds both, so the exclusion radius can be stated for either record without choosing between them. Jordan’s inequality $q(k) \geq (4/\pi^2)|k|^2$ also holds on the whole zone and is tight at $k = 0$ and at the corner, but the corner is not where this bound is used; on the axis it is loose by $\pi^2/4$, and the constants it gives, 17.04 and 23.66, are exactly $\pi/2$ times the sharp ones. It is not a bound on the open coefficient: a value of C_{shp} outside the two records carries its own W_4 , and nothing here confines the answer to the interval the records span. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and that threshold is the same for both constants — the sharp radius reaches π when $\frac{u}{2}\sqrt{W_4/(\theta t_3)} = 1$, which is exactly where $Ku = \pi$ for the Jordan constant, because the two bounds agree at the zone corner. The excluded zone fraction grows as u^3 .

check: Jordan bound $q(k) \geq (4/\pi^2)|k|^2$ holds on the whole zone

check: the exact zone minimum of q on $|k| \geq r$ is $4 \sin^2(r/2)$, and Jordan overstates it by $\pi/2$

check: the criterion survives C2: K depends on the kernel only through $\sqrt{W_4}$

check: the statement is non-vacuous only below an explicit coupling

10 External comparisons

External results count here because they are *independent* — produced without knowledge of this program — not because they are published. Each is an assertion until something checks it.

Hamer 1989, both sectors. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2, 4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12} . The 0^{++} series matches the charge-even A_1^{++} Γ -point coefficients at x^2 and x^3 , with the sign structure ($a_1 = -1$ in the scalar channel where the carrier’s is $+1$) reproduced; the E^{++} doublet is not constrained by it, which is why Proposition 24 needs the splitting and not the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer’s normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [2]; that paper has not been read here, so the disagreement is recorded and not adjudicated.

check: Hamer’s 1^{+-} series matches the C-odd Gamma-point coefficients through x^3

check: Hamer’s 0^{++} series matches the C-even Gamma-point coefficients through x^3

check: the $m_n = 2^{n-1}a_n$ bridge is the $x = 2u$ conversion

check: the Hamer table is pinned, and the a.4 agreement is primary-source

The string tension. The Kogut–Pearson–Shigemitsu $SU(3)$ table [4] — read here from the 1980 preprint, the journal pages themselves being unread — equals this program’s certified σ series exactly, $2^{n-1}t_n = \sigma_n$ for $n = 2, \dots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3 , including the discriminating u^3 coefficient.

check: the KPS 1980 string-tension table equals the certified sigma series EXACTLY

check: $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$, and C5 is the $n = 3$ case

check: the ratio and sigma series reproduce E_{flat} exactly

Euclidean series. Two printings of the same Münster/Seo series — the second erratum to [5], which prints the definitive corrected table and credits Ukawa and Seo [6] with locating the error, and Smit’s Table 1 scalar column [7], which credits them — agree rational for rational through u^8 . That is agreement transcription for transcription, not two independent computations, and it is recorded as such. One shift is recorded and not adjudicated: Münster’s 1985 table [9] differs from the 1982 erratum row at eighth order by exactly -96 , a fourth correction already in print by 1983 [8] that no erratum records. This is a Euclidean record; no Hamiltonian comparison is drawn from it, and none is permitted.

check: the errata-resolved Euclidean series is doubly sourced, transcription for transcription

check: FINDING: Münster’s 1985 table shifts his 1982 erratum at eighth order

The overlap obstruction. Schierholz [10] measured in 1988 that at $\beta = 5.9$ — in the Euclidean scaling region, not at strong coupling — a bare local operator carries only a couple of percent of the state it interpolates, and gave the scaling a^5 for a local dimension-four operator. Nothing is transferred from that measurement to here: it is a Euclidean Monte Carlo result at the opposite end of the coupling axis from $u \ll 1$, and the firewall of Section 11 forbids importing it as a bound. Read as a warning rather than as a bound, it is why G18 must be posed in a smeared basis: an overlap that

falls as a power of the spacing is not something better statistics recover.

check: the overlap obstruction was published in 1988, and it scales

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [17], whose Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [14]. The rule is enforced by the literature validator rather than recorded in prose.

check: a cross-regime paper never supplies a value

11 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-} ; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L , but its nearest incidence separation is $\Delta_L(u) = 4\tau(u) \sin^2(\pi/L) + O(u^4)$, which falls as L^{-2} and so provides no volume-uniform spectral isolation.

check: $\Delta_L = 4 \tau(u) \sin^2(\pi/L)$ is positive and falls as L^{-2}

Three crossings are named explicitly, and none of them is made here.

- **Finite lattice to infinite volume** requires an inhomogeneous Wilson free-energy bound with a useful $\log K_\alpha$ and a source-radius reduction. Both are open, and are carried in the accompanying gap register as G17.
- **Protected level to particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the carrier with the transfer-matrix spectrum, and multi-plaquette survival of the protected level (G18). The published overlap measurement of Section 10 is why this theorem must live in the smeared basis.
- **Lattice to continuum** (G19) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

The second and third are the only route from the statements above to any sentence containing the words “mass gap”, and nothing in this edition shortens that route.

Four premises earlier in the paper are arguments in prose rather than results, and all four are load-bearing. None has a check.

That the range of P_- is exactly the charge-odd one-plaquette span for $L \geq 3$ is argued in Section 2 from the link count of a reduced fundamental loop.

That one-shared-link processes exhaust the second-order *off-diagonal* intermediates is assumed in Theorem 10. Three neighbouring statements *are* settled and should not be confused with it: the channel list is complete (Lemma 6), two distinct faces of this lattice share at most one link, and the vacuum-mediated route vanishes in this sector (Proposition 12). The gap is in the process enumeration, not in the channel list, and Theorem 7 is silent about it.

That the degree-(2,2) Haar moment over $SU(N)$ equals the $U(N)$ one is argued in the proof of Theorem 7 from the $U(1)$ invariance of the integrand and the exhaustion of the invariants below the rank. The argument is standard and the range is the theorem’s own, but nothing here checks it.

That the charge-even sector’s second-order Bloch symbol is the unsigned incidence $\mathcal{N}(k)$ — the signed incidence with the boundary orientations dropped — is asserted in Section 4 and used throughout Section 6 and Appendix D. The registry records it as a convention rather than a result, and the closed-form cubic, the range, the bandwidth and the Γ -splitting all rest on it.

12 Reproducibility

A portable standard-library Python program, `paper/verify_core.py` in the accompanying repository (the root-level file of the same name belongs to the earlier flat-band manuscript and is a different, smaller program), runs twenty-five checks in nine claim groups: the rational coefficient ledger; the Weingarten derivation of Theorem 7 by explicit index summation; the incidence identity, *decided* in exact Gaussian-rational arithmetic at rational torus points rather than checked to a residual; the chain condition $\partial_2 \partial_3 = 0$ over $\mathbb{Z}$ and the finite-torus ranks with an explicit kernel basis, together with the closed cube surface of Proposition 19; the Rayleigh quotient of the wrapping sheets; the two-cube connected geometry rebuilt from oriented cell boundaries, its exact $B = 6$ and $B = 4$ spectra, the endpoint Casimir budgets and the weak retention rule they decide, and Reported result 14 channel by channel; the charge-even cubic of Theorem 21 with $p + q = 12$ and the four high-symmetry spectra of both sectors and the Γ splitting of Proposition 24; the checkpoint algebra of (23), the second witness, and the

vanishing of e_2 on every axial cut; and the coupling convention, including the 16 and 64 by which a 4^r rescaling misses. Every value in it is a **Fraction**: no float is constructed anywhere, so “agreement” there always means equality. It has no dependencies and runs in about half a second.

Nothing in `verify_core.py` assumes or decides a value of C_{shp} . Its fourth-order group is the obstruction as arithmetic: the extraction formulas, the blindness of X , and the identical vanishing of e_2 on axial cuts.

Every theorem, proposition, corollary and reported result above names the machine check that establishes it, and the appendix tables name checks too. Three displayed objects deliberately carry none, because none is a result of this program: the dimension and Casimir table (10), which is standard $SU(N)$ representation theory; the boundary formulas of Appendix E, which are definitions — what is checked there is that they compose to zero; and Table 1, which is provenance about a commit rather than a claim about the physics, and whose rows are re-measured by the commands printed beside it. The checks live in the accompanying repository, where `workhouse verify --only '<name>'` re-runs any one of them alone, with its numbers, in about a second. That device is itself checked: a registered invariant reads every pinned edition in `paper/` — this one included — resolves every printed label against the live registry, and fails if one is renamed, deleted, or left red. The guard is the reason this edition’s labels can be trusted at all: the same invariant read only the previous edition until this one landed, and a label that had drifted passed unnoticed until it was made to read both.

This edition pins commit `af3ab5f2e9860e75372c1a143a33c828f8b0d9ba` of the accompanying repository — the commit at which every check name printed above resolves, and the last one before this file itself lands. It carries the counters in Table 1. They describe *that* commit and no other, and each row has its own command: after `git checkout af3ab5f`, `make verify` gives the T_1 and T_2 rows and `make check` the test count, while the T_0 row is the declaration scrape that `workhouse certified` prints — `make lean` compiles the tree and reports no count. They are provenance rather than independent evidence; the argument is above, and the checks are the checks.

A smaller layer is proof-checked rather than merely re-derived. The Lean development compiles the rank law with denominators cleared, $t_3 = 5/612$ and $t_2 = 0$, the third-order ledger identity, the axial law at every exceptional rank, the cycle count $(L^3 - 1) + 3 = L^3 + 2$, the four checkpoint-extraction formulas and the deltas they invert, and four statements about the charge-even cubic including the identity that its middle coefficient is p^2 . It is exact-rational and polynomial algebra, plus the trigonometric checkpoint evaluations over $\mathbb{R}$ that supply those deltas — the $\mathbb{R}$ section exists because the

Layer	Count
T_0 — Lean 4 theorems, no <code>sorry</code>	40
T_1 — exact re-derivations	188
T_2 — numerical, within a stated tolerance	49
Repository tests	486

Table 1: Counters at the pinned commit. The T_0 row counts theorem declarations with no `sorry`, which is a scrape of the tree; `make lean` is what compiles them, and the standard the tree is held to is that every axiom footprint lies in `{propxt, Classical.choice, Quot.sound}`. That compilation was last run elsewhere: the environment this edition was typeset in carries no Lean toolchain, and the row is not asserted as re-measured here.

extraction theorems formalised only half a statement without it. There is no perturbative derivation and no operator theory.

An earlier edition added that “nothing that bears on C_{shp} appears there”. That was false, and false about the one genuinely open item here. The Lean core defines the historical side $C_{\text{old}} = -211835444920651/4405310420659200$ as an exact rational and proves three internal-consistency relations about it: C from β , the width as $\alpha + \beta$, and the tier-collapse relation. What it contains no spelling of, anywhere in the tree, is the v10a.26 value — so the layer touches C_{shp} and still prefers neither side, and that even-handedness is an absence checked by inspection rather than asserted.

check: the Lean core DOES carry the historical C_{shp} side, and still adjudicates nothing

Machine checks certify algebra relative to stated definitions and inputs; they do not replace a derivation, and they do not certify an uncompleted off-axis contraction — which is why Section 9 stops where it does.

A Channel weights in closed form

Substituting (10) into (12),

$$w_1 = -\frac{2}{N(N^2 - 1)}, \quad w_{\Lambda^2 F} = -\frac{N - 1}{(N + 1)(2N - 3)},$$

$$w_{\text{Adj}} = -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, \quad w_{\text{Sym}^2 F} = -\frac{N + 1}{(N - 1)(2N + 3)}.$$

Their family sums are (13) and (14), and the difference has numerator $2N(N^2 - 4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once.

check: the four channel weights follow from dimension and Casimir

At $N = 3$: $w_1 = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\text{Sym}^2 F} = -2/9$, which are the four values Reported result 14 matches against (19).

B The Weingarten index sums

With $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$, and σ, τ running over S_2 ,

$$\begin{aligned} & \langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle \\ &= \sum_{\sigma, \tau} \text{Wg}(\sigma \tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}. \end{aligned}$$

Summing the delta products over all N^4 index quadruples,

$$M_{\text{direct}} = \text{Wg}(e)(N^4 + N^2) + 2 \text{Wg}((12))N^3 = N^2,$$

$$M_{\text{cross}} = 2 \text{Wg}(e)N^3 + \text{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$.

check: the shared-link weights are Weingarten, not an isotropy assumption

The related fourth moment $\int |U_{11}|^4 dU = 2/[N(N + 1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

check: $SU(3)$ Weingarten values follow from the general formula

check: the fourth moment integral $|U_{11}|^4 = 1/6$ at $N = 3$

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

check: the manuscript's $SU(3)$ ledger is this registry, value by value

The u^1 row is $SU(3)$ -only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, - \rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. Of the rest, the u^0 row and the u^2 $BB^\dagger$ coefficient specialise all-rank statements ($2C_F$ and t_N); the u^2 flat scalar inherits the $SU(3)$ tower term through (20); and the three u^3 rows are $SU(3)$ -only and conditional on the supplied domino ledger, exactly as Section 7 says.

D The charge-even ledger

In the same convention, with $\lambda \in \{12, -4\}$ and (20). The $\lambda = -4$ rows and the bandwidth read off them are even- L , thermodynamic-limit values: $\lambda = -4$ requires some $k_j = \pi$, which the periodic grid samples only at even L . The $\lambda = 12$ rows sit at Γ and hold on every grid.

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16 t_+ $	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	on-site diagonal ($\lambda = 0$)	$52163/260100$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

check: one assembly formula gives every C-even value at both orders

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it **bandmin** at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

check: FINDING: the certificate key 'bandmin' holds the band MAXIMUM, at both orders

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned} \partial_3[x; 1, 2, 3] &= [x + e_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + e_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + e_3; 1, 2] - [x; 1, 2]. \end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2 \partial_3 = 0$ over $\mathbb{Z}$.

check: d.2 d.3 = 0 on the built complex

On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$.

check: rank d.3 = L^3 - 1 on the built complex

F The two-cube geometry

The eleven oriented plaquettes of the $(3, 2, 2)$ prism, by base vertex and ordered plane axes, are

$$\begin{aligned} &(000; 12), (000; 13), (000; 23), (001; 12), \\ &(010; 13), (100; 12), (100; 13), (100; 23), \\ &(101; 12), (110; 13), (200; 23), \end{aligned}$$

with left, right and shared-face index sets $I_L = (0, 1, 2, 3, 4, 7)$, $I_R = (5, 6, 7, 8, 9, 10)$, $I_F = (7)$. Building $G = BB^\top$ from the oriented boundaries and applying the geometric Möbius fold (17) leaves exactly four nonzero off-diagonal pairs, $(0, 5)$, $(1, 6)$, $(3, 8)$, $(4, 9)$, each equal to -1 : four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form once the diagonals below are supplied. The two connected diagonals are

$$\begin{aligned} D_{B6} &= \frac{1}{612} \text{diag}(-2317, -2317, -2295, -2317, -2317, \\ &\quad -2317, -2317, 0, -2317, -2317, -2295), \\ D_{B4} &= \text{diag}\left(-\frac{7}{4}, -\frac{7}{4}, -\frac{15}{4}, -\frac{7}{4}, -\frac{7}{4}, \right. \\ &\quad \left. -\frac{7}{4}, -\frac{7}{4}, 0, -\frac{7}{4}, -\frac{7}{4}, -\frac{15}{4}\right), \end{aligned}$$

so both spectra are one line of arithmetic: a pair block contributes $D_{ii} \mp c$ twice and each isolated direction its own D_{ii} . The $B = 4$ comparator on the same geometry gives $\{0, (-\frac{15}{4})^2, (-\frac{11}{6})^4, (-\frac{5}{3})^4\}$: the paired levels sit *above* the isolated $-15/4$ pair, where at $B = 6$ they sit below it. The three isolated levels $\{-\frac{15}{4}, -\frac{15}{4}, 0\}$ are identical in the two truncations, so the entire effect of the cutoff lives in the four cross-cell blocks. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \rightarrow -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

check: the B=4 comparator on the same geometry gives
-1/12 with the reversed spectrum

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